即 $\alpha_1\Delta y+\alpha_2\Delta x=o(\rho)$，故 $f(x,y)$ 在点 (x_0,y_0) 处可微.

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注 类似地，可以证明

若 $f'_x(x,y)$，$f'_y(x,y)$ 都在点 (x_0,y_0) 处连续，则 $f(x,y)$ 在点 (x_0,y_0) 处可微.

第二节　偏导数与全微分的计算

一、考试内容要点精讲

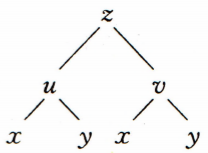
深度精读

1. 复合函数求导法

设 $u=u(x,y)$，$v=v(x,y)$ 可导，$z=f(u,v)$ 在相应点有连续一阶偏导数，则

$$\frac{\partial z}{\partial x}=\frac{\partial f}{\partial u}\frac{\partial u}{\partial x}+\frac{\partial f}{\partial v}\frac{\partial v}{\partial x},$$

$$\frac{\partial z}{\partial y}=\frac{\partial f}{\partial u}\frac{\partial u}{\partial y}+\frac{\partial f}{\partial v}\frac{\partial v}{\partial y}.$$



2. 全微分形式不变性

设 $z=f(u,v)$，$u=u(x,y)$，$v=v(x,y)$ 都有连续一阶偏导数. 则

$$\mathrm{d}z=\frac{\partial z}{\partial x}\mathrm{d}x+\frac{\partial z}{\partial y}\mathrm{d}y,\mathrm{d}z=\frac{\partial z}{\partial u}\mathrm{d}u+\frac{\partial z}{\partial v}\mathrm{d}v.$$

3. 隐函数求导法

(1) 由一个方程所确定的隐函数

设 $F(x,y,z)$ 有连续一阶偏导数，$F'_z\neq 0$，$z=z(x,y)$ 由 $F(x,y,z)=0$ 所确定.

方法

① 公式：$\frac{\partial z}{\partial x}=-\frac{F'_x}{F'_z},\frac{\partial z}{\partial y}=-\frac{F'_y}{F'_z}$；

② 等式两边求导：$F'_x+F'_z\frac{\partial z}{\partial x}=0,F'_y+F'_z\frac{\partial z}{\partial y}=0$；

③ 利用微分形式不变性：$F'_x\mathrm{d}x+F'_y\mathrm{d}y+F'_z\mathrm{d}z=0$.

(2) 由方程组所确定的隐函数

设 $u=u(x,y)$，$v=v(x,y)$ 由 $\begin{cases}F(x,y,u,v)=0\\G(x,y,u,v)=0\end{cases}$ 所确定.

方法

① 等式两边求导 $\begin{cases}F'_x+F'_u\frac{\partial u}{\partial x}+F'_v\frac{\partial v}{\partial x}=0,\\G'_x+G'_u\frac{\partial u}{\partial x}+G'_v\frac{\partial v}{\partial x}=0;\end{cases}$

② 利用微分形式不变性 $\begin{cases}F'_x\mathrm{d}x+F'_y\mathrm{d}y+F'_u\mathrm{d}u+F'_v\mathrm{d}v=0,\\G'_x\mathrm{d}x+G'_y\mathrm{d}y+G'_u\mathrm{d}u+G'_v\mathrm{d}v=0.\end{cases}$

二、常考题型的方法与技巧

题型一　求一点处的偏导数与全微分

展示新知

【例 1】 设 $f(x,y)=\begin{cases}\dfrac{\sqrt{|x|}}{x^2+y^2}\sin(x^2+y^2), & (x,y)\neq(0,0),\\ 0, & (x,y)=(0,0).\end{cases}$ 求 $f'_x(0,0)$ 和 $f'_y(0,0)$.

解 由于 $\lim\limits_{\Delta x\to 0}\dfrac{f(\Delta x,0)-f(0,0)}{\Delta x}=\lim\limits_{\Delta x\to 0}\dfrac{\frac{\sqrt{|\Delta x|}}{(\Delta x)^2}\sin(\Delta x)^2}{\Delta x}=\lim\limits_{\Delta x\to 0}\dfrac{\sqrt{|\Delta x|}}{\Delta x}=\infty$,

则 $f'_x(0,0)$ 不存在. 而

$$f'_y(0,0)=\lim_{\Delta y\to 0}\frac{f(0,\Delta y)-f(0,0)}{\Delta y}=\lim_{\Delta y\to 0}\frac{0-0}{\Delta y}=0.$$

注 求分段函数在分界点处的偏导数一般都是用定义.

展示新知

【例 2】 设 $f(x,y)=\dfrac{2x+3y}{1+xy\sqrt{x^2+y^2}}$,求 $f'_x(0,0)$ 和 $f'_y(0,0)$.

解
$$f'_x(0,0)=\frac{\mathrm{d}}{\mathrm{d}x}f(x,0)\Big|_{x=0}=\frac{\mathrm{d}}{\mathrm{d}x}(2x)\Big|_{x=0}=2,$$

$$f'_y(0,0)=\frac{\mathrm{d}}{\mathrm{d}y}f(0,y)\Big|_{y=0}=\frac{\mathrm{d}}{\mathrm{d}y}(3y)\Big|_{y=0}=3.$$

注 求具体点处的偏导数可用先代后求的方法往往比较简单.

展示新知

【例 3】 设 $z=\ln(1+xy^2)$,则 $\dfrac{\partial^2 z}{\partial x\partial y}\Big|_{(0,1)}=$ _____.

解 $\dfrac{\partial z}{\partial x}=\dfrac{y^2}{1+xy^2}$,$\dfrac{\partial^2 z}{\partial x\partial y}\Big|_{(0,1)}=\dfrac{\mathrm{d}}{\mathrm{d}y}\left(\dfrac{\partial z(0,y)}{\partial x}\right)\Big|_{y=1}=\dfrac{\mathrm{d}}{\mathrm{d}y}(y^2)\Big|_{y=1}=2.$

注 求具体点处的高阶偏导数也可用先代后求的方法.

展示新知

【例 4】 设 $f(x,y,z)=\sqrt[z]{\dfrac{x}{y}}$,则 $\mathrm{d}f(1,1,1)=$ _____.

解
$$f'_x(1,1,1)=\frac{\mathrm{d}}{\mathrm{d}x}f(x,1,1)\Big|_{x=1}=\frac{\mathrm{d}}{\mathrm{d}x}(x)\Big|_{x=1}=1,$$

$$f'_y(1,1,1)=\frac{\mathrm{d}}{\mathrm{d}y}f(1,y,1)\Big|_{y=1}=\frac{\mathrm{d}}{\mathrm{d}y}\left(\frac{1}{y}\right)\Big|_{y=1}=-\frac{1}{y^2}\Big|_{y=1}=-1,$$

$$f'_z(1,1,1)=\frac{\mathrm{d}}{\mathrm{d}z}f(1,1,z)=\frac{\mathrm{d}}{\mathrm{d}z}(1)\Big|_{z=1}=0,$$

故 $\mathrm{d}f(1,1,1)=\mathrm{d}x-\mathrm{d}y$.

题型二　求已给出具体表达式函数的偏导数与全微分

训练思维

【例 1】　设 $z=(x^2+y^2)e^{-\arctan\frac{y}{x}}$，求 $\frac{\partial z}{\partial x}$，$\frac{\partial z}{\partial y}$ 及 dz.

解　$$\frac{\partial z}{\partial x}=(2x+y)e^{-\arctan\frac{y}{x}},\frac{\partial z}{\partial y}=(2y-x)e^{-\arctan\frac{y}{x}},$$

$$dz=(2x+y)e^{-\arctan\frac{y}{x}}dx+(2y-x)e^{-\arctan\frac{y}{x}}dy.$$

训练思维

【例 2】　设 $z=(1+x^2+y^2)^{xy}$，求 $\frac{\partial z}{\partial x}$ 及 $\frac{\partial z}{\partial y}$.

解　（方法一）　由原题设可知 $z=e^{xy\ln(1+x^2+y^2)}$，两端对 x,y 分别求偏导.

（方法二）　由原题设知 $\ln z=xy\ln(1+x^2+y^2)$，两端对 x,y 分别求偏导.

（方法三）　令 $u=1+x^2+y^2,v=xy$，则函数可看作 $z=u^v$，由复合函数求导法可知

$$\frac{\partial z}{\partial x}=\frac{\partial z}{\partial u}\frac{\partial u}{\partial x}+\frac{\partial z}{\partial v}\frac{\partial v}{\partial x}=vu^{v-1}2x+u^v\ln u\cdot y$$

$$=(1+x^2+y^2)^{xy}\left[\frac{2x^2y}{1+x^2+y^2}+y\ln(1+x^2+y^2)\right].$$

同理可得　$\frac{\partial z}{\partial y}=(1+x^2+y^2)^{xy}\left[x\ln(1+x^2+y^2)+\frac{2xy^2}{1+x^2+y^2}\right]$.

注　方法三也可用于一元幂指函数，如 $y=(1+x^2)^{\sin x}$，可令 $u=1+x^2,v=\sin x$.

训练思维

【例 3】　若函数 $z=f(x,y)$ 满足 $\frac{\partial^2 z}{\partial y^2}=2$，且 $f(x,1)=x+2$，又 $f'_y(x,1)=x+1$，则 $f(x,y)$ 等于

(A) $y^2+(x-1)y-2$.　　(B) $y^2+(x+1)y+2$.

(C) $y^2+(x-1)y+2$.　　(D) $y^2+(x+1)y-2$.

解　（方法一）　容易验证，只有(C)选项中的函数同时满足题设中的三个条件，故应选(C).

（方法二）　由 $\frac{\partial^2 z}{\partial y^2}=2$ 知 $\frac{\partial z}{\partial y}=\int 2dy=2y+\varphi(x)$. 由题设条件 $f'_y(x,1)=1+x$ 知，

$$1+x=2+\varphi(x)\Rightarrow\varphi(x)=x-1\Rightarrow\frac{\partial z}{\partial y}=2y+x-1,$$

于是 $z=\int(2y+x-1)dy=y^2+y(x-1)+\psi(x)$.

由 $f(x,1)=x+2$ 知 $x+2=1+(x-1)+\psi(x)$，从而 $\psi(x)=2$.

则 $z=y^2+y(x-1)+2$. 故应选(C).

注　方法一只适用于选择题，方法二是一般方法.

训练思维

【例 4】　已知 $\frac{\partial^2 z}{\partial x\partial y}=1$，且当 $x=0$ 时，$z=\sin y$；当 $y=0$ 时，$z=\sin x$，则 $z(x,y)=$_____.

解 （方法一） 由$\frac{\partial^2 z}{\partial x \partial y}=1$知$\frac{\partial z}{\partial x}=\int 1\mathrm{d}y=y+\varphi(x)$. 于是

$$z=\int[y+\varphi(x)]\mathrm{d}x=xy+\int\varphi(x)\mathrm{d}x+\psi(y)=xy+g(x)+\psi(y),$$

其中$g(x)=\int\varphi(x)\mathrm{d}x$.

由$x=0$时，$z=\sin y$知

$$\sin y=g(0)+\psi(y). \tag{①}$$

由$y=0$时，$z=\sin x$知

$$\sin x=g(x)+\psi(0).$$

从而有$z=xy+\sin x+\sin y-g(0)-\psi(0)$.

在①式中令$y=0$得$g(0)+\psi(0)=0$，故$z(x,y)=xy+\sin x+\sin y$.

（方法二） 由$\frac{\partial^2 z}{\partial x \partial y}=1$知$\frac{\partial z}{\partial x}=\int 1\mathrm{d}y=y+\varphi(x)$.

又当$y=0$时，$z=\sin x$，则$z(x,0)=\sin x$，$z'_x(x,0)=\cos x$，从而有

$$\varphi(x)=\cos x,\frac{\partial z}{\partial x}=y+\cos x,$$

即

$$z=\int(y+\cos x)\mathrm{d}x=xy+\sin x+\psi(y).$$

由$x=0$时，$z=\sin y$知，$\psi(y)=\sin y$. 故$z(x,y)=xy+\sin x+\sin y$.

训练思维

【例5】(仅数一) 已知$(axy^3-y^2\cos x)\mathrm{d}x+(1+by\sin x+3x^2y^2)\mathrm{d}y$是某一函数的全微分，则$a,b$取值分别为

(A) -2和2. (B)2和-2. (C) -3和3. (D)3和-3.

解 由题设可知，存在可微函数$f(x,y)$，使

$$\mathrm{d}f(x,y)=(axy^3-y^2\cos x)\mathrm{d}x+(1+by\sin x+3x^2y^2)\mathrm{d}y,$$

则

$$\frac{\partial f}{\partial x}=axy^3-y^2\cos x,\frac{\partial f}{\partial y}=1+by\sin x+3x^2y^2,$$

从而有

$$\frac{\partial^2 f}{\partial x\partial y}=3axy^2-2y\cos x,\frac{\partial^2 f}{\partial y\partial x}=by\cos x+6xy^2.$$

由于$\frac{\partial^2 f}{\partial x\partial y}$和$\frac{\partial^2 f}{\partial y\partial x}$都连续，从而有$\frac{\partial^2 f}{\partial x\partial y}=\frac{\partial^2 f}{\partial y\partial x}$，即

$$3axy^2-2y\cos x=by\cos x+6xy^2,$$

则$\begin{cases}3a=6,\\ b=-2,\end{cases}$即$\begin{cases}a=2,\\ b=-2.\end{cases}$故应选(B).

注 由本题求解中可看出，若$P(x,y)$，$Q(x,y)$有连续一阶偏导数，且$P(x,y)\mathrm{d}x+Q(x,y)\mathrm{d}y$是某一函数全微分，则$\frac{\partial P}{\partial y}=\frac{\partial Q}{\partial x}$. 此结论以后可直接用，本题直接用该结论更简单.

训练思维

【例 6】 设 $f(x)$ 有连续一阶导数，且有 $[xy-yf(x)]\mathrm{d}x+[f(x)+y^2]\mathrm{d}y=\mathrm{d}u(x,y)$，求 $f(x)$ 及 $u(x,y)$，其中 $f(0)=-1$.

解 由题设知 $x-f(x)=f'(x)$，即

$$f'(x)+f(x)=x,$$

$$f(x)=(x-1)+Ce^{-x}.$$

由 $f(0)=-1$ 知，$C=0$，$f(x)=x-1$，从而

$$\mathrm{d}u(x,y)=y\mathrm{d}x+[(x-1)+y^2]\mathrm{d}y.$$

（方法一） 偏积分 由上式可知 $\dfrac{\partial u}{\partial x}=y$，$\dfrac{\partial u}{\partial y}=x-1+y^2$.

等式 $\dfrac{\partial u}{\partial x}=y$ 两端对 x 偏积分得

$$u=\int y\mathrm{d}x=xy+\varphi(y),$$

该式两端对 y 求偏导得

$$\frac{\partial u}{\partial y}=x+\varphi'(y).$$

又

$$\frac{\partial u}{\partial y}=x-1+y^2,$$

则 $\varphi'(y)=-1+y^2$，从而

$$\varphi(y)=\int(-1+y^2)\mathrm{d}y=-y+\frac{1}{3}y^3+C,$$

$$u(x,y)=xy-y+\frac{1}{3}y^3+C.$$

（方法二） 凑微分

$$\begin{aligned}\mathrm{d}u(x,y)&=y\mathrm{d}x+[(x-1)+y^2]\mathrm{d}y\\&=(y\mathrm{d}x+x\mathrm{d}y)+(y^2-1)\mathrm{d}y\\&=\mathrm{d}(xy)+\mathrm{d}\left(\frac{1}{3}y^3-y\right)\\&=\mathrm{d}\left(xy+\frac{1}{3}y^3-y\right),\end{aligned}$$

则 $u(x,y)=xy-y+\dfrac{1}{3}y^3+C$.

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注 类似地，

已知函数 $f(x,y)$ 满足 $\mathrm{d}f(x,y)=\dfrac{x\mathrm{d}y-y\mathrm{d}x}{x^2+y^2}$，$f(1,1)=\dfrac{\pi}{4}$，则 $f(\sqrt{3},3)=$ _____.

$\left(\dfrac{\pi}{3}\right)$

题型三　含有抽象函数的复合函数偏导数与全微分

训练思维

【例 1】 设函数 $f(x,y)$ 可微，且 $f(x+1,e^x)=x(x+1)^2$，$f(x,x^2)=2x^2\ln x$，则 $df(1,1)=$

(A)$dx+dy$.　(B)$dx-dy$.　(C)dy.　(D)$-dy$.

解 （方法一） 直接法

等式 $f(x+1,e^x)=x(x+1)^2$ 两端对 x 求导得

$$f_1'(x+1,e^x)+f_2'(x+1,e^x)e^x=(x+1)^2+2x(x+1),$$

上式中令 $x=0$ 得

$$f_1'(1,1)+f_2'(1,1)=1.$$

等式 $f(x,x^2)=2x^2\ln x$ 两端对 x 求导得

$$f_1'(x,x^2)+2xf_2'(x,x^2)=4x\ln x+2x,$$

上式中令 $x=1$ 得

$$f_1'(1,1)+2f_2'(1,1)=2,$$

从而可得 $f_1'(1,1)=0$，$f_2'(1,1)=1$，则 $df(1,1)=dy$. 故应选(C).

（方法二） 排除法

令 $x+1=u$，$e^x=v$，由 $f(x+1,e^x)=x(x+1)^2$ 得 $f(u,v)=u^2\ln v$. 且满足 $f(x,x^2)=2x^2\ln x$，则

$$f_u'(1,1)=0,f_v'(1,1)=1,$$

$df(1,1)=dy$. 则排除(A)(B)(D)，故应选(C).

训练思维

【例 2】 设函数 $u(x,y)=\varphi(x+y)+\varphi(x-y)+\int_{x-y}^{x+y}\psi(t)dt$，其中 φ 具有二阶导数，ψ 具有一阶导数，则必有

(A) $\frac{\partial^2 u}{\partial x^2}=-\frac{\partial^2 u}{\partial y^2}$.　(B) $\frac{\partial^2 u}{\partial x^2}=\frac{\partial^2 u}{\partial y^2}$.　(C) $\frac{\partial^2 u}{\partial x\partial y}=\frac{\partial^2 u}{\partial y^2}$.　(D) $\frac{\partial^2 u}{\partial x\partial y}=-\frac{\partial^2 u}{\partial x^2}$.

解 （方法一） 直接法

$$\frac{\partial u}{\partial x}=\varphi'(x+y)+\varphi'(x-y)+\psi(x+y)-\psi(x-y),$$

$$\frac{\partial u}{\partial y}=\varphi'(x+y)-\varphi'(x-y)+\psi(x+y)+\psi(x-y),$$

进一步有：

$$\frac{\partial^2 u}{\partial x^2}=\varphi''(x+y)+\varphi''(x-y)+\psi'(x+y)-\psi'(x-y),$$

$$\frac{\partial^2 u}{\partial y^2}=\varphi''(x+y)+\varphi''(x-y)+\psi'(x+y)-\psi'(x-y),$$

从而有 $\frac{\partial^2 u}{\partial x^2}=\frac{\partial^2 u}{\partial y^2}$. 故应选(B).

（方法二） 排除法

令 $\varphi(x)=x^2$，$\psi(x)=0$，则

$$u(x,y)=(x+y)^2+(x-y)^2=2x^2+2y^2,$$

$$\frac{\partial^2 u}{\partial x^2}=4,\frac{\partial^2 u}{\partial y^2}=4,\frac{\partial^2 u}{\partial x\partial y}=0.$$

显然(A)(C)(D)均不正确,故应选(B).

训练思维

【例3】 设 $z=f(xy,x^2+y^2)$,求 $\frac{\partial z}{\partial x},\frac{\partial^2 z}{\partial x\partial y}$,其中 $f(u,v)$ 有二阶连续偏导数.

解 $\frac{\partial z}{\partial x}=yf'_1+2xf'_2$,

$$\frac{\partial^2 z}{\partial x\partial y}=f'_1+y(xf''_{11}+2yf''_{12})+2x(f''_{21}x+f''_{22}\cdot 2y)$$

$$=f'_1+xy(f''_{11}+4f''_{22})+2(x^2+y^2)f''_{12}.$$

训练思维

【例4】 设 $f(x,y)$ 可微,又 $f(0,0)=0,f'_x(0,0)=a,f'_y(0,0)=b$ 且 $g(t)=f[t,f(t,t^2)]$,求 $g'(0)$.

解 $g'(t)=f'_1[t,f(t,t^2)]+f'_2[t,f(t,t^2)]\cdot[f'_1(t,t^2)+f'_2(t,t^2)\cdot 2t]$,

$$g'(0)=f'_1(0,0)+f'_2(0,0)[f'_1(0,0)+f'_2(0,0)\times 0]$$

$$=a+b(a+0\times b)=a(1+b).$$

训练思维

【例5】 设 $u=f(x,y,z),y=\varphi(x,t),t=\psi(x,z)$,其中 f,φ,ψ 可微,求 $\frac{\partial u}{\partial x},\frac{\partial u}{\partial z}$.

解 $\frac{\partial u}{\partial x}=\frac{\partial f}{\partial x}+\frac{\partial f}{\partial y}\left(\frac{\partial \varphi}{\partial x}+\frac{\partial \varphi}{\partial t}\frac{\partial \psi}{\partial x}\right)$.

$$\frac{\partial u}{\partial z}=\frac{\partial f}{\partial y}\frac{\partial \varphi}{\partial t}\frac{\partial \psi}{\partial z}+\frac{\partial f}{\partial z}.$$

训练思维

【例6】 设 $f(u,v)$ 具有二阶连续偏导数,且满足 $\frac{\partial^2 f}{\partial u^2}+\frac{\partial^2 f}{\partial v^2}=1$,又 $g(x,y)=f[xy,\frac{1}{2}(x^2-y^2)]$,求 $\frac{\partial^2 g}{\partial x^2}+\frac{\partial^2 g}{\partial y^2}$.

解 令 $xy=u,\frac{1}{2}(x^2-y^2)=v$,则

$$\frac{\partial g}{\partial x}=\frac{\partial f}{\partial u}y+\frac{\partial f}{\partial v}x,$$

$$\frac{\partial^2 g}{\partial x^2}=y\left(\frac{\partial^2 f}{\partial u^2}y+\frac{\partial^2 f}{\partial u\partial v}x\right)+x\left(\frac{\partial^2 f}{\partial v\partial u}y+\frac{\partial^2 f}{\partial v^2}x\right)+\frac{\partial f}{\partial v},$$

$$\frac{\partial g}{\partial y}=\frac{\partial f}{\partial u}x-\frac{\partial f}{\partial v}y,$$

$$\frac{\partial^2 g}{\partial y^2}=x\left(\frac{\partial^2 f}{\partial u^2}x-\frac{\partial^2 f}{\partial u\partial v}y\right)-y\left(\frac{\partial^2 f}{\partial v\partial u}x-\frac{\partial^2 f}{\partial v^2}y\right)-\frac{\partial f}{\partial v},$$

故 $\frac{\partial^2 g}{\partial x^2}+\frac{\partial^2 g}{\partial y^2}=(x^2+y^2)\left(\frac{\partial^2 f}{\partial u^2}+\frac{\partial^2 f}{\partial v^2}\right)=x^2+y^2.$

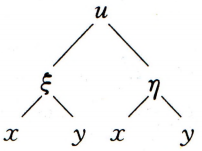
训练思维

【例 7】 设函数 $u=f(x,y)$ 具有二阶连续偏导数，且满足

$$4\frac{\partial^2 u}{\partial x^2}+12\frac{\partial^2 u}{\partial x\partial y}+5\frac{\partial^2 u}{\partial y^2}=0,$$

确定 a,b 的值，使等式在变换 $\xi=x+ay,\eta=x+by$ 下简化为 $\frac{\partial^2 u}{\partial\xi\partial\eta}=0$.

解 （方法一） 变量之间的关系如右图



$$\frac{\partial u}{\partial x}=\frac{\partial u}{\partial\xi}\cdot 1+\frac{\partial u}{\partial\eta}\cdot 1=\frac{\partial u}{\partial\xi}+\frac{\partial u}{\partial\eta},$$

$$\begin{aligned}\frac{\partial^2 u}{\partial x^2}&=\frac{\partial^2 u}{\partial\xi^2}\cdot 1+\frac{\partial^2 u}{\partial\xi\partial\eta}\cdot 1+\frac{\partial^2 u}{\partial\eta\partial\xi}\cdot 1+\frac{\partial^2 u}{\partial\eta^2}\cdot 1\\&=\frac{\partial^2 u}{\partial\xi^2}+2\frac{\partial^2 u}{\partial\xi\partial\eta}+\frac{\partial^2 u}{\partial\eta^2},\end{aligned}$$

$$\frac{\partial u}{\partial y}=\frac{\partial u}{\partial\xi}\cdot a+\frac{\partial u}{\partial\eta}\cdot b=a\frac{\partial u}{\partial\xi}+b\frac{\partial u}{\partial\eta},$$

$$\begin{aligned}\frac{\partial^2 u}{\partial y^2}&=a\frac{\partial^2 u}{\partial\xi^2}\cdot a+a\frac{\partial^2 u}{\partial\xi\partial\eta}\cdot b+b\frac{\partial^2 u}{\partial\eta\partial\xi}\cdot a+b\frac{\partial^2 u}{\partial\eta^2}\cdot b\\&=a^2\frac{\partial^2 u}{\partial\xi^2}+2ab\frac{\partial^2 u}{\partial\xi\partial\eta}+b^2\frac{\partial^2 u}{\partial\eta^2},\end{aligned}$$

$$\begin{aligned}\frac{\partial^2 u}{\partial x\partial y}&=\frac{\partial^2 u}{\partial\xi^2}\cdot a+\frac{\partial^2 u}{\partial\xi\partial\eta}\cdot b+\frac{\partial^2 u}{\partial\eta\partial\xi}\cdot a+\frac{\partial^2 u}{\partial\eta^2}\cdot b\\&=a\frac{\partial^2 u}{\partial\xi^2}+(a+b)\frac{\partial^2 u}{\partial\xi\partial\eta}+b\frac{\partial^2 u}{\partial\eta^2}.\end{aligned}$$

将以上三个二阶偏导数代入等式 $4\frac{\partial^2 u}{\partial x^2}+12\frac{\partial^2 u}{\partial x\partial y}+5\frac{\partial^2 u}{\partial y^2}=0$ 得

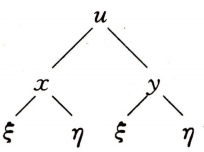
$$(5a^2+12a+4)\frac{\partial^2 u}{\partial\xi^2}+[10ab+12(a+b)+8]\frac{\partial^2 u}{\partial\xi\partial\eta}+(5b^2+12b+4)\frac{\partial^2 u}{\partial\eta^2}=0.$$

由题设知 $\begin{cases}5a^2+12a+4=0,\\5b^2+12b+4=0,\end{cases}$ 但 $10ab+12(a+b)+8\neq 0$，解得

$$\begin{cases}a=-2,\\b=-\frac{2}{5}\end{cases}\text{或}\begin{cases}a=-\frac{2}{5},\\b=-2.\end{cases}$$

（方法二） 由 $\xi=x+ay,\eta=x+by$ 解得

$$\begin{cases}x=\frac{a\eta-b\xi}{a-b},\\y=\frac{\xi-\eta}{a-b}.\end{cases}$$



$$\frac{\partial u}{\partial\xi}=\frac{-b}{a-b}\frac{\partial u}{\partial x}+\frac{1}{a-b}\frac{\partial u}{\partial y},$$

$$\frac{\partial^2 u}{\partial\xi\partial\eta}=\frac{-b}{a-b}\frac{\partial^2 u}{\partial x^2}\cdot\frac{a}{a-b}+\frac{-b}{a-b}\frac{\partial^2 u}{\partial x\partial y}\cdot\frac{-1}{a-b}+\frac{1}{a-b}\frac{\partial^2 u}{\partial y\partial x}\cdot\frac{a}{a-b}+\frac{1}{a-b}\frac{\partial^2 u}{\partial y^2}\cdot\frac{-1}{a-b}$$

$$=\frac{-ab}{(a-b)^2}\frac{\partial^2 u}{\partial x^2}+\frac{a+b}{(a-b)^2}\frac{\partial^2 u}{\partial x\partial y}+\frac{-1}{(a-b)^2}\frac{\partial^2 u}{\partial y^2}.$$

欲使$\frac{\partial^2 u}{\partial \xi\partial \eta}=0$，即

$$-ab\frac{\partial^2 u}{\partial x^2}+(a+b)\frac{\partial^2 u}{\partial x\partial y}-\frac{\partial^2 u}{\partial y^2}=0.$$

与已知关系式比较得 $$\frac{-ab}{4}=\frac{a+b}{12}=\frac{-1}{5},$$

由此解得$\begin{cases}a=-2,\\ b=-\frac{2}{5}\end{cases}$或$\begin{cases}a=-\frac{2}{5},\\ b=-2.\end{cases}$

解析扫封面二维码获取

注 类似地，

已知函数 $f(u,v)$ 具有二阶连续偏导数，且函数 $g(x,y)=f(2x+y,3x-y)$ 满足

$$\frac{\partial^2 g}{\partial x^2}+\frac{\partial^2 g}{\partial x\partial y}-6\frac{\partial^2 g}{\partial y^2}=1.$$

（Ⅰ）求$\frac{\partial^2 f}{\partial u\partial v}$；$\left(\frac{1}{25}\right)$

（Ⅱ）若$\frac{\partial f(u,0)}{\partial u}=ue^{-u}$，$f(0,v)=\frac{1}{50}v^2-1$，求 $f(u,v)$ 的表达式.

$\left(\frac{1}{25}uv-e^{-u}(u+1)+\frac{1}{50}v^2\right)$

训练思维

【例 8】 设 $f(u)$ 具有二阶连续导数，而 $z=f(e^x\sin y)$ 满足方程$\frac{\partial^2 z}{\partial x^2}+\frac{\partial^2 z}{\partial y^2}=ze^{2x}$，求 $f(u)$.

解 令 $u=e^x\sin y$，则

$$\frac{\partial z}{\partial x}=f'(u)e^x\sin y,\frac{\partial^2 z}{\partial x^2}=f''(u)e^{2x}\sin^2 y+f'(u)e^x\sin y,$$

$$\frac{\partial z}{\partial y}=f'(u)e^x\cos y,\frac{\partial^2 z}{\partial y^2}=f''(u)e^{2x}\cos^2 y-f'(u)e^x\sin y.$$

将$\frac{\partial^2 z}{\partial x^2}$和$\frac{\partial^2 z}{\partial y^2}$代入等式$\frac{\partial^2 z}{\partial x^2}+\frac{\partial^2 z}{\partial y^2}=ze^{2x}$得

$$f''(u)=f(u),\text{即 } f''(u)-f(u)=0.$$

这是一个二阶线性常系数齐次微分方程，特征方程为 $r^2-1=0$，$r=\pm 1$，则

$$f(u)=C_1e^u+C_2e^{-u}.$$

训练思维

【例 9】 设 (r,θ) 为极坐标，$u=u(r,\theta)$ 具有二阶连续偏导数，并满足$\frac{\partial u}{\partial \theta}\equiv 0$，且$\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0$，求 $u(r,\theta)$.

解 由$\frac{\partial u}{\partial \theta}\equiv 0$知 u 仅为 r 的函数，令 $u=\varphi(r)$，其中 $r=\sqrt{x^2+y^2}$，则

$$\frac{\partial u}{\partial x}=\varphi'(r)\frac{x}{\sqrt{x^2+y^2}}=\varphi'(r)\frac{x}{r},$$

$$\frac{\partial^2 u}{\partial x^2}=\varphi''(r)\frac{x^2}{r^2}+\varphi'(r)\frac{r-\frac{x^2}{r}}{r^2}=\varphi''(r)\frac{x^2}{r^2}+\varphi'(r)\left(\frac{1}{r}-\frac{x^2}{r^3}\right).$$

由对称性知

$$\frac{\partial^2 u}{\partial y^2}=\varphi''(r)\frac{y^2}{r^2}+\varphi'(r)\left(\frac{1}{r}-\frac{y^2}{r^3}\right),$$

则

$$\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=\varphi''(r)+\varphi'(r)\frac{1}{r}.$$

从而得 $\varphi''(r)+\varphi'(r)\frac{1}{r}=0$，即 $r\varphi''(r)+\varphi'(r)=0$，从而$[r\varphi'(r)]'=0$. 则 $r\varphi'(r)=C_1$，$\varphi(r)=C_1\ln r+C_2$. 故 $u=C_1\ln r+C_2$.

解析扫封面二维码获取

注 类似地，

(仅数一) 设 $f(x)$ 在$[1,+\infty)$上二阶连续可导，$f(1)=0$，$f'(1)=1$. $z=(x^2+y^2)f(x^2+y^2)$ 满足$\frac{\partial^2 z}{\partial x^2}+\frac{\partial^2 z}{\partial y^2}=0$，求 $f(x)$ 的表达式.

$\left(f(x)=\frac{\ln x}{x}\right)$

训练思维

【例 10】 若对任意 $t>0$ 有 $f(tx,ty)=t^n f(x,y)$，则称函数 $f(x,y)$ 是 n 次齐次函数，试证：若 $f(x,y)$ 可微，则 $f(x,y)$ 是 n 次齐次函数 $\Leftrightarrow x\frac{\partial f}{\partial x}+y\frac{\partial f}{\partial y}=nf(x,y)$.

证明 **必要性** 由于 $f(x,y)$ 为 n 次齐次函数，则对任意 $t>0$，有 $f(tx,ty)=t^n f(x,y)$，该式两端对 t 求导得

$$xf'_1(tx,ty)+yf'_2(tx,ty)=nt^{n-1}f(x,y).$$

令 $t=1$ 得

$$xf'_1(x,y)+yf'_2(x,y)=nf(x,y),$$

即 $x\frac{\partial f}{\partial x}+y\frac{\partial f}{\partial y}=nf(x,y)$.

充分性 令 $F(t)=f(tx,ty)(t>0)$，则

$$\frac{\mathrm{d}F}{\mathrm{d}t}=xf'_1(tx,ty)+yf'_2(tx,ty),$$

两边同乘 t 得

$$t\frac{\mathrm{d}F}{\mathrm{d}t}=txf'_1(tx,ty)+tyf'_2(tx,ty)=nf(tx,ty)=nF(t),$$

于是$\frac{\mathrm{d}F}{F}=\frac{n}{t}\mathrm{d}t$，解得 $F(t)=Ct^n$.

令 $t=1$ 得 $F(1)=C$，而由 $F(t)=f(tx,ty)$ 知$F(1)=f(x,y)$，则 $C=f(x,y)$.

于是 $F(t)=t^n f(x,y)$，即 $f(tx,ty)=t^n f(x,y)$.

解析扫封面二维码获取

注 利用本题结论解下列题目更简单.

设 $f(u)$ 可导，$z=xyf\left(\frac{y}{x}\right)$，若 $x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=xy(\ln y-\ln x)$，则

(A) $f(1)=\frac{1}{2}, f'(1)=0$.　　(B) $f(1)=0, f'(1)=\frac{1}{2}$.

(C) $f(1)=\frac{1}{2}, f'(1)=1$.　　(D) $f(1)=0, f'(1)=1$.　　(B)

题型四　隐函数的偏导数与全微分

展示新知

【例 1】 设 $z=z(x,y)$ 由方程 $x-az=e^{y+az}$（a 是非零常数）确定，则

(A) $\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}=\frac{1}{a}$.　(B) $\frac{\partial z}{\partial x}+\frac{\partial z}{\partial y}=\frac{1}{a}$.　(C) $\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}=-\frac{1}{a}$.　(D) $\frac{\partial z}{\partial x}+\frac{\partial z}{\partial y}=-\frac{1}{a}$.

解 （方法一） 令 $F(x,y)=x-az-e^{y+az}$，

则 $F'_x=1, F'_y=-e^{y+az}, F'_z=-a-ae^{y+az}=-a(1+e^{y+az})$.

$$\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}=-\frac{F'_x}{F'_z}+\frac{F'_y}{F'_z}=\frac{1}{a(1+e^{y+az})}+\frac{e^{y+az}}{a(1+e^{y+az})}=\frac{1}{a}.$$

故选(A).

（方法二） 等式 $x-az=e^{y+az}$ 两边同时对 x 求偏导得

$$1-a\frac{\partial z}{\partial x}=e^{y+az}\left(a\frac{\partial z}{\partial x}\right), \frac{\partial z}{\partial x}=\frac{1}{a(1+e^{y+az})}.$$

等式 $x-az=e^{y+az}$ 两边同时对 y 求偏导得

$$-a\frac{\partial z}{\partial y}=e^{y+az}(1+a\frac{\partial z}{\partial y}), \frac{\partial z}{\partial y}=-\frac{e^{y+az}}{a(1+e^{y+az})}.$$

$$\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}=\frac{1}{a(1+e^{y+az})}+\frac{e^{y+az}}{a(1+e^{y+az})}=\frac{1}{a}.$$

（方法三） 等式 $x-az=e^{y+az}$ 两端微分得

$$dx-adz=e^{y+az}(dy+adz).$$

$$dz=\frac{1}{a(1+e^{y+az})}dx-\frac{e^{y+az}}{a(1+e^{y+az})}dy.$$

$$\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}=\frac{1}{a(1+e^{y+az})}+\frac{e^{y+az}}{a(1+e^{y+az})}=\frac{1}{a}.$$

解析扫封面二维码获取

注 类似地，

设函数 $z=z(x,y)$ 由方程 $z+e^x-y\ln(1+z^2)=0$ 确定，求 $\left(\frac{\partial^2 z}{\partial x^2}+\frac{\partial^2 z}{\partial y^2}\right)\Big|_{(0,0)}$.

$(-1-2\ln 2)$

展示新知

【例 2】 设方程 $F\left(\frac{x}{z},\frac{z}{y}\right)=0$ 可确定函数 $z=z(x,y)$，求 $\frac{\partial z}{\partial x}$ 和 $\frac{\partial z}{\partial y}$.

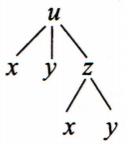
解 由隐函数求导公式得

$$\frac{\partial z}{\partial x}=-\frac{\frac{1}{z}F_1'}{-\frac{x}{z^2}F_1'+\frac{1}{y}F_2'}=\frac{yzF_1'}{xyF_1'-z^2F_2'},$$

$$\frac{\partial z}{\partial y}=-\frac{-\frac{z}{y^2}F_2'}{-\frac{x}{z^2}F_1'+\frac{1}{y}F_2'}=\frac{-z^3F_2'}{y(xyF_1'-z^2F_2')}.$$

展示新知

【例 3】 设 $u=f(x,y,z)$ 有连续一阶偏导数，$z=z(x,y)$ 由方程 $xe^x-ye^y=ze^z$ 所确定，求 du.



解 （方法一） 由题设知

$$\frac{\partial u}{\partial x}=\frac{\partial f}{\partial x}+\frac{\partial f}{\partial z}\frac{\partial z}{\partial x},$$

等式 $xe^x-ye^y=ze^z$ 两端对 x 求导得

$$e^x+xe^x=(e^z+ze^z)\frac{\partial z}{\partial x}.$$

由此可得

$$\frac{\partial z}{\partial x}=\frac{e^x(1+x)}{e^z(1+z)}=\frac{1+x}{1+z}e^{x-z},$$

则

$$\frac{\partial u}{\partial x}=\frac{\partial f}{\partial x}+\frac{\partial f}{\partial z}\frac{1+x}{1+z}e^{x-z}.$$

同理可求得

$$\frac{\partial u}{\partial y}=\frac{\partial f}{\partial y}-\frac{\partial f}{\partial z}\frac{1+y}{1+z}e^{y-z}.$$

故 $$du=\left(\frac{\partial f}{\partial x}+\frac{\partial f}{\partial z}\frac{1+x}{1+z}e^{x-z}\right)dx+\left(\frac{\partial f}{\partial y}-\frac{\partial f}{\partial z}\frac{1+y}{1+z}e^{y-z}\right)dy.$$

（方法二） 根据微分形式不变性，由 $u=f(x,y,z)$ 知，$du=\frac{\partial f}{\partial x}dx+\frac{\partial f}{\partial y}dy+\frac{\partial f}{\partial z}dz$.

等式 $xe^x-ye^y=ze^z$ 两端求微分得

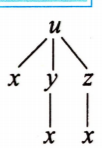
$$(e^x+xe^x)dx-(e^y+ye^y)dy=(e^z+ze^z)dz,$$

解得 $dz=\frac{1+x}{1+z}e^{x-z}dx-\frac{1+y}{1+z}e^{y-z}dy$. 将 dz 代入 $du=\frac{\partial f}{\partial x}dx+\frac{\partial f}{\partial y}dy+\frac{\partial f}{\partial z}dz$ 得

$$du=\left(\frac{\partial f}{\partial x}+\frac{\partial f}{\partial z}\frac{1+x}{1+z}e^{x-z}\right)dx+\left(\frac{\partial f}{\partial y}-\frac{\partial f}{\partial z}\frac{1+y}{1+z}e^{y-z}\right)dy.$$

训练思维

【例 4】 设 $u=f(x,y,z)$，$\varphi(x^2,e^y,z)=0$，$y=\sin x$ 确定了函数 $u=u(x)$，其中 f,φ 都有一阶连续偏导数，且 $\frac{\partial\varphi}{\partial z}\neq 0$，求 $\frac{du}{dx}$.



解 （方法一） $\frac{du}{dx}=\frac{\partial f}{\partial x}+\frac{\partial f}{\partial y}\cos x+\frac{\partial f}{\partial z}\frac{dz}{dx}$.

$\varphi(x^2,e^y,z)=0$ 两端对 x 求导得

$$\varphi_1'2x+\varphi_2'e^y\cos x+\varphi_3'\frac{dz}{dx}=0,$$

解得 $\frac{dz}{dx}=-\frac{1}{\varphi_3'}(2x\varphi_1'+\varphi_2'e^y\cos x)$. 将 $\frac{dz}{dx}$ 代入 $\frac{du}{dx}=\frac{\partial f}{\partial x}+\frac{\partial f}{\partial y}\cos x+\frac{\partial f}{\partial z}\frac{dz}{dx}$ 得

$$\frac{\mathrm{d}u}{\mathrm{d}x}=\frac{\partial f}{\partial x}+\frac{\partial f}{\partial y}\cos x-\frac{\dfrac{\partial f}{\partial z}}{\varphi_3'}(2x\varphi_1'+\varphi_2'\mathrm{e}^y\cos x).$$

（方法二） 根据微分形式不变性，由 $u=f(x,y,z)$ 知

$$\mathrm{d}u=\frac{\partial f}{\partial x}\mathrm{d}x+\frac{\partial f}{\partial y}\mathrm{d}y+\frac{\partial f}{\partial z}\mathrm{d}z, \tag{1}$$

等式 $\varphi(x^2,\mathrm{e}^y,z)=0$ 两端求微分得

$$\varphi_1'2x\mathrm{d}x+\varphi_2'\mathrm{e}^y\mathrm{d}y+\varphi_3'\mathrm{d}z=0. \tag{2}$$

由 $y=\sin x$ 知 $\mathrm{d}y=\cos x\mathrm{d}x$，将 $\mathrm{d}y=\cos x\mathrm{d}x$ 代入 ② 式得

$$\mathrm{d}z=-\frac{1}{\varphi_3'}(\varphi_1'2x+\varphi_2'\mathrm{e}^y\cos x)\mathrm{d}x.$$

将该式中的 $\mathrm{d}z$ 和 $\mathrm{d}y=\cos x\mathrm{d}x$ 代入 ① 式得

$$\mathrm{d}u=\left[\frac{\partial f}{\partial x}+\frac{\partial f}{\partial y}\cos x-\frac{\dfrac{\partial f}{\partial z}}{\varphi_3'}(2x\varphi_1'+\varphi_2'\mathrm{e}^y\cos x)\right]\mathrm{d}x.$$

故 $\dfrac{\mathrm{d}u}{\mathrm{d}x}=\dfrac{\partial f}{\partial x}+\dfrac{\partial f}{\partial y}\cos x-\dfrac{\dfrac{\partial f}{\partial z}}{\varphi_3'}(2x\varphi_1'+\varphi_2'\mathrm{e}^y\cos x)$.

训练思维

【例 5】 设 $y=f(x,t)$，且方程 $F(x,y,t)=0$ 确定了函数 $t=t(x,y)$，其中 f,F 都具有一阶连续偏导数，求 $\dfrac{\mathrm{d}y}{\mathrm{d}x}$.

解 **（方法一）** 将 $t=t(x,y)$ 代入 $y=f(x,t)$ 得 $y=f(x,t(x,y))$，这是一个关于 x,y 的二元方程，它可确定 y 是 x 的函数.

等式 $y=f(x,t(x,y))$ 两端对 x 求导得

$$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{\partial f}{\partial x}+\frac{\partial f}{\partial t}\left(\frac{\partial t}{\partial x}+\frac{\partial t}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}x}\right),$$

而 $t=t(x,y)$ 由 $F(x,y,t)=0$ 所确定，则

$$\frac{\partial t}{\partial x}=-\frac{\dfrac{\partial F}{\partial x}}{\dfrac{\partial F}{\partial t}},\frac{\partial t}{\partial y}=-\frac{\dfrac{\partial F}{\partial y}}{\dfrac{\partial F}{\partial t}},$$

于是 $\dfrac{\mathrm{d}y}{\mathrm{d}x}=\dfrac{\partial f}{\partial x}-\dfrac{\partial f}{\partial t}\left(\dfrac{\dfrac{\partial F}{\partial x}}{\dfrac{\partial F}{\partial t}}+\dfrac{\dfrac{\partial F}{\partial y}}{\dfrac{\partial F}{\partial t}}\dfrac{\mathrm{d}y}{\mathrm{d}x}\right)\Rightarrow\dfrac{\mathrm{d}y}{\mathrm{d}x}=\dfrac{\dfrac{\partial F}{\partial t}\dfrac{\partial f}{\partial x}-\dfrac{\partial F}{\partial x}\dfrac{\partial f}{\partial t}}{\dfrac{\partial F}{\partial t}+\dfrac{\partial F}{\partial y}\dfrac{\partial f}{\partial t}}$.

（方法二） 根据微分形式不变性，由 $y=f(x,t)$ 知 $\mathrm{d}y=\dfrac{\partial f}{\partial x}\mathrm{d}x+\dfrac{\partial f}{\partial t}\mathrm{d}t$.

由 $F(x,y,t)=0$ 知 $\dfrac{\partial F}{\partial x}\mathrm{d}x+\dfrac{\partial F}{\partial y}\mathrm{d}y+\dfrac{\partial F}{\partial t}\mathrm{d}t=0$.

解得 $\mathrm{d}t=-\dfrac{1}{\dfrac{\partial F}{\partial t}}\left(\dfrac{\partial F}{\partial x}\mathrm{d}x+\dfrac{\partial F}{\partial y}\mathrm{d}y\right)$. 将 $\mathrm{d}t$ 的表达式代入 $\mathrm{d}y=\dfrac{\partial f}{\partial x}\mathrm{d}x+\dfrac{\partial f}{\partial t}\mathrm{d}t$ 并整理可得

$$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{\dfrac{\partial F}{\partial t}\dfrac{\partial f}{\partial x}-\dfrac{\partial F}{\partial x}\dfrac{\partial f}{\partial t}}{\dfrac{\partial F}{\partial t}+\dfrac{\partial F}{\partial y}\dfrac{\partial f}{\partial t}}.$$